

Guide to the Seller A Best-Response Plots

1 Model Objects Used in the Plots

The state in the simulation is the user's history with Seller A,

$$(S, F),$$

where S is the number of successes previously observed from Seller A and F is the number of failures. Because the user does not observe Seller A's hidden product choice, the user updates a misspecified single-product belief,

$$\theta \mid S, F \sim \text{Beta}(1 + S, 1 + F).$$

The probability that the Thompson-sampling user chooses Seller A is

$$\rho(S, F) = \Pr(\tilde{\theta} \geq p_0 \mid S, F), \quad \tilde{\theta} \sim \text{Beta}(1 + S, 1 + F),$$

where p_0 is the known success probability of Seller B.

For product $x \in \{1, 2\}$, define Seller A's product-specific action value as

$$Q_x(S, F) = R - c_x + \gamma [p_x V(S + 1, F) + (1 - p_x) V(S, F + 1)].$$

The optimal product at state (S, F) is the product with the larger Q value. Equivalently, product 2 is optimal when

$$Q_2(S, F) - Q_1(S, F) \geq 0.$$

Since

$$Q_2(S, F) - Q_1(S, F) = -(c_2 - c_1) + \gamma(p_2 - p_1) [V(S + 1, F) - V(S, F + 1)],$$

product 2 is optimal exactly when

$$D(S, F) \equiv V(S + 1, F) - V(S, F + 1) \geq \frac{c_2 - c_1}{\gamma(p_2 - p_1)}.$$

The value-difference plots visualize this threshold condition directly.

2 Best Response as p_0 Varies

Figure 1 summarizes how Seller A's incentives change with p_0 .

- The top-left panel plots $Q_2(0, 0) - Q_1(0, 0)$ at the initial state. Positive values mean product 2 is optimal at the prior.
- The top-right panel compares initial demand for A with simulated market share. Initial demand is high when p_0 is low and falls as Seller B becomes stronger.

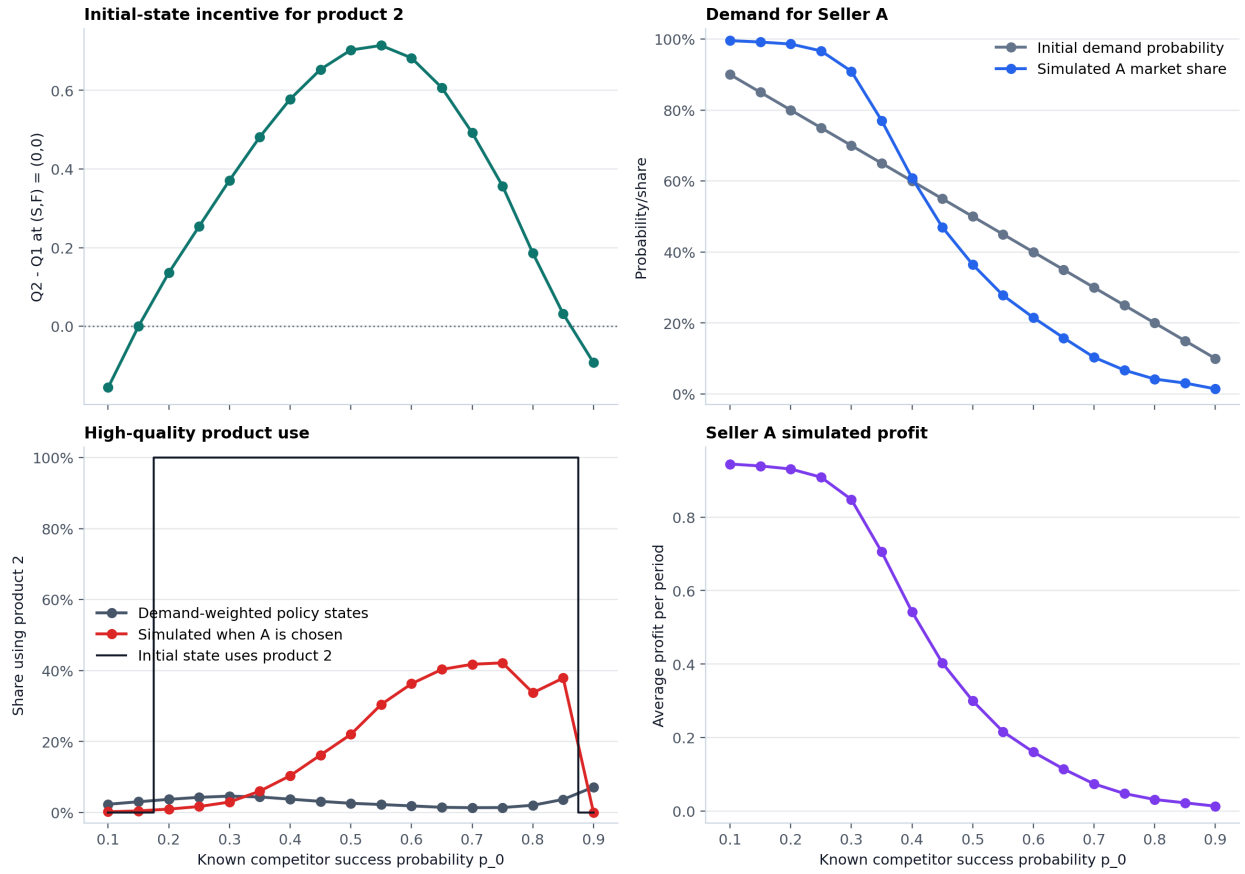


Figure 1: Summary of Seller A's best response as Seller B's known success probability p_0 varies.

- The bottom-left panel compares product 2 use across solved policy states and simulated histories.
- The bottom-right panel shows that Seller A's average profit falls as p_0 increases, because A is chosen less often and must sometimes pay for the more expensive product to protect reputation.

The key pattern is hump-shaped: product 2 is most attractive when Seller A is neither safely ahead nor hopelessly behind. If p_0 is very low, Seller B is weak, so Seller A can often save cost with product 1. If p_0 is very high, the chance of future demand from A is too low to justify much investment. For intermediate p_0 , a success can meaningfully improve future demand, so product 2 is more valuable.

3 Policy Heatmaps in Count Space

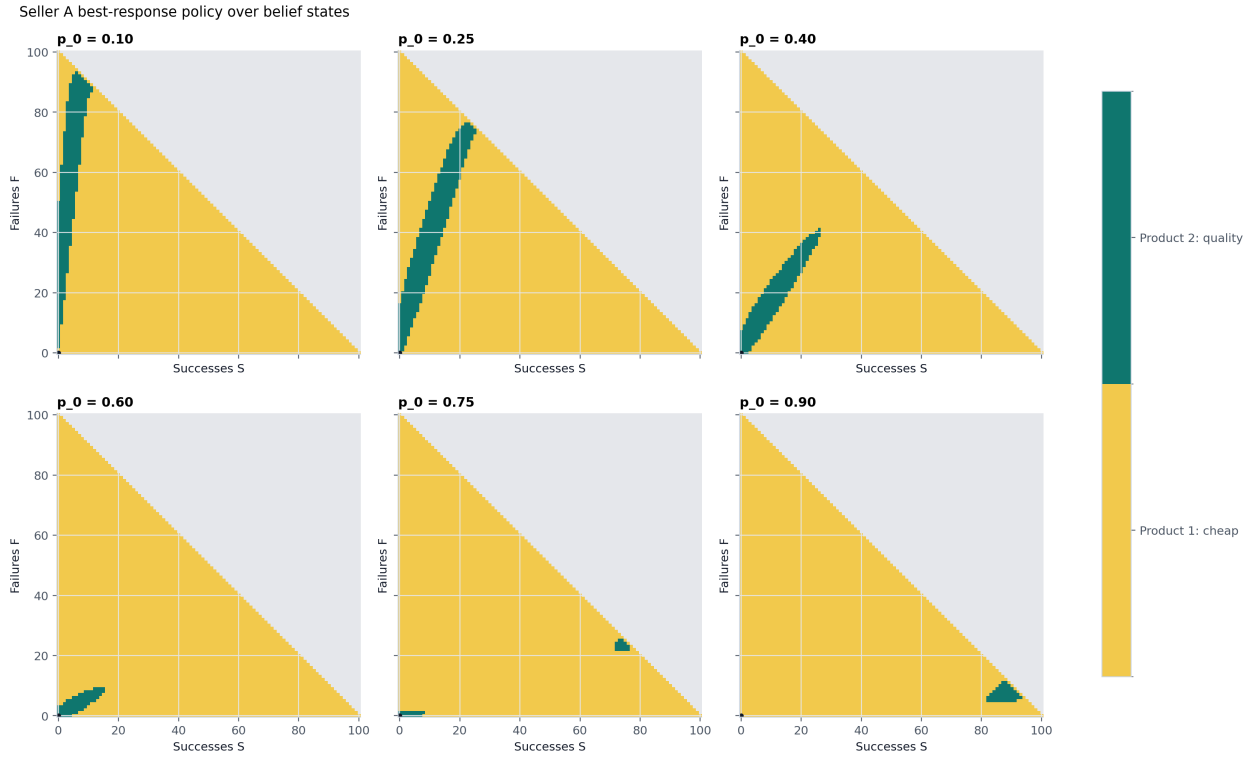


Figure 2: Optimal product choice over the raw count state space (S, F) .

Figure 2 plots the optimal product for Seller A over the state space (S, F) .

- The horizontal axis is the number of observed successes S .
- The vertical axis is the number of observed failures F .
- Yellow means product 1, the cheap product.
- Green means product 2, the high-quality product.

- The black dot is the initial state $(0,0)$.
- The gray region is infeasible under the truncation $S + F \leq N$.

The green regions are states where a success is sufficiently valuable for future demand. These regions are not simply “bad reputation” or “good reputation” states. Instead, they tend to occur near reputation margins where the next outcome can move the user’s belief enough to change future choice probabilities.

4 Policy Over Posterior State Space

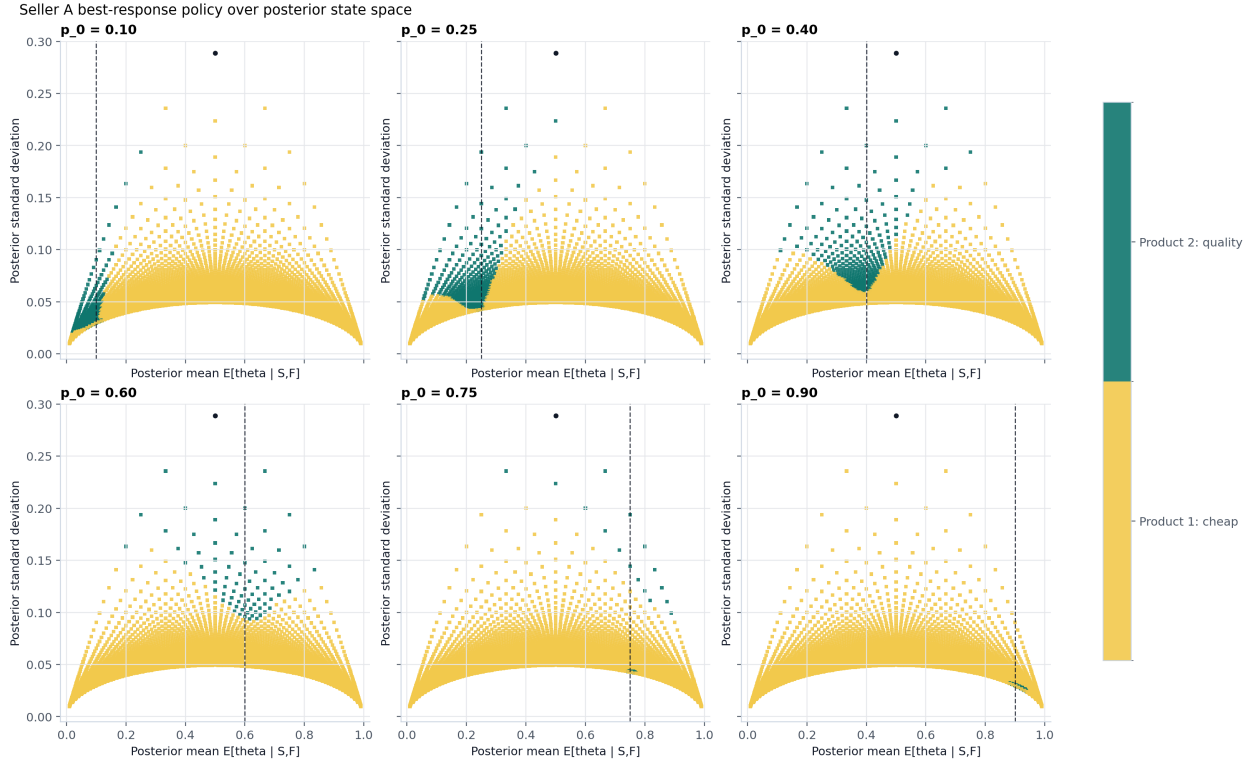


Figure 3: Optimal product choice over posterior mean and posterior uncertainty.

Figure 3 re-expresses the same policy in posterior belief coordinates. Each point is still a count state (S, F) , but it is plotted using

$$m(S, F) = \mathbb{E}[\theta \mid S, F] = \frac{1 + S}{2 + S + F}$$

on the horizontal axis and the posterior standard deviation on the vertical axis.

The dashed vertical line is p_0 . It is the known success probability of Seller B and therefore the Thompson-sampling comparison threshold. The black dot is the initial prior $\text{Beta}(1, 1)$, which has posterior mean 0.5 and standard deviation $\sqrt{1/12}$.

The fan shape is mechanical. As $S + F$ grows, the posterior becomes more precise, so points move downward. At the same time, different success rates generate different posterior means, spreading points horizontally. The upper part of the fan corresponds to low-count, high-uncertainty states; the lower part corresponds to high-count, precise states.

5 Value-Difference Diagnostic

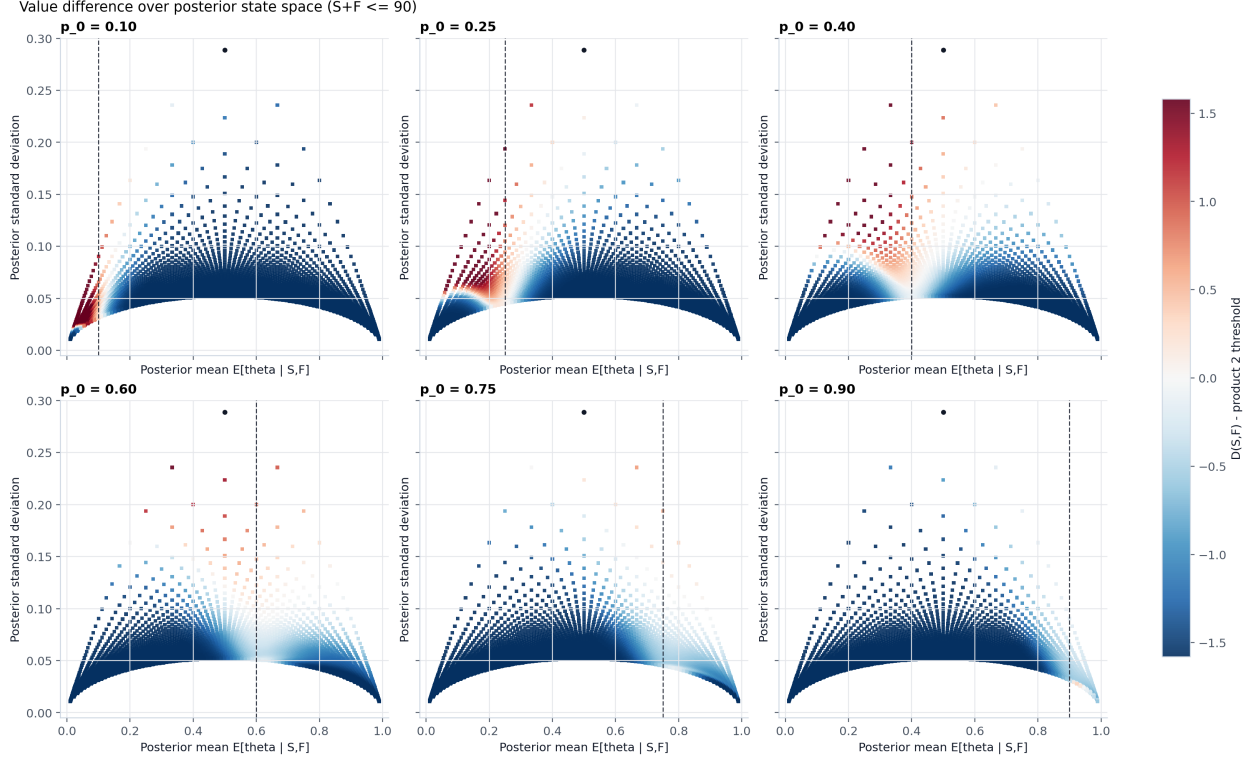


Figure 4: Interior diagnostic for the value-difference threshold $D(S, F)$.

Figure 4 plots

$$D(S, F) - \frac{c_2 - c_1}{\gamma(p_2 - p_1)}.$$

Positive values mean product 2 is optimal, while negative values mean product 1 is optimal. This plot is useful because it shows the smooth object behind the policy rule. The policy plot discretizes states into yellow and green, while the value-difference plot shows whether those regions come from a coherent threshold or from numerical noise.

The displayed version uses the interior restriction $S + F \leq 90$, while the baseline value iteration solves the grid up to $N = 100$. This removes the top boundary band and helps check whether the visible policy pattern is driven by the artificial truncation.

6 Truncation Robustness

The infinite state space is approximated by solving states with $S + F \leq N$. The baseline plots use $N = 100$. To check whether the boundary condition is driving the policy, the code re-solves the model at $N = 150$ and $N = 200$ and compares policy choices on the common state space.

The main conclusion is that the policy is very stable away from the truncation edge. Disagreements are concentrated near the artificial boundary $S + F = N$. The high-uncertainty points in the posterior-state plots are not boundary states: high posterior uncertainty comes from low $S + F$, whereas the truncation boundary is at high $S + F$.

Table 1: Policy disagreement when comparing the baseline $N = 100$ solution to larger truncations.

Comparison N	States compared	Max disagreement	Mean disagreement
150	All common states, $S + F \leq 100$	1.0095%	0.3163%
150	Safe interior, $S + F \leq 90$	0.1911%	0.0576%
200	All common states, $S + F \leq 100$	1.0095%	0.3163%
200	Safe interior, $S + F \leq 90$	0.1911%	0.0576%

7 Simulation Path Plot

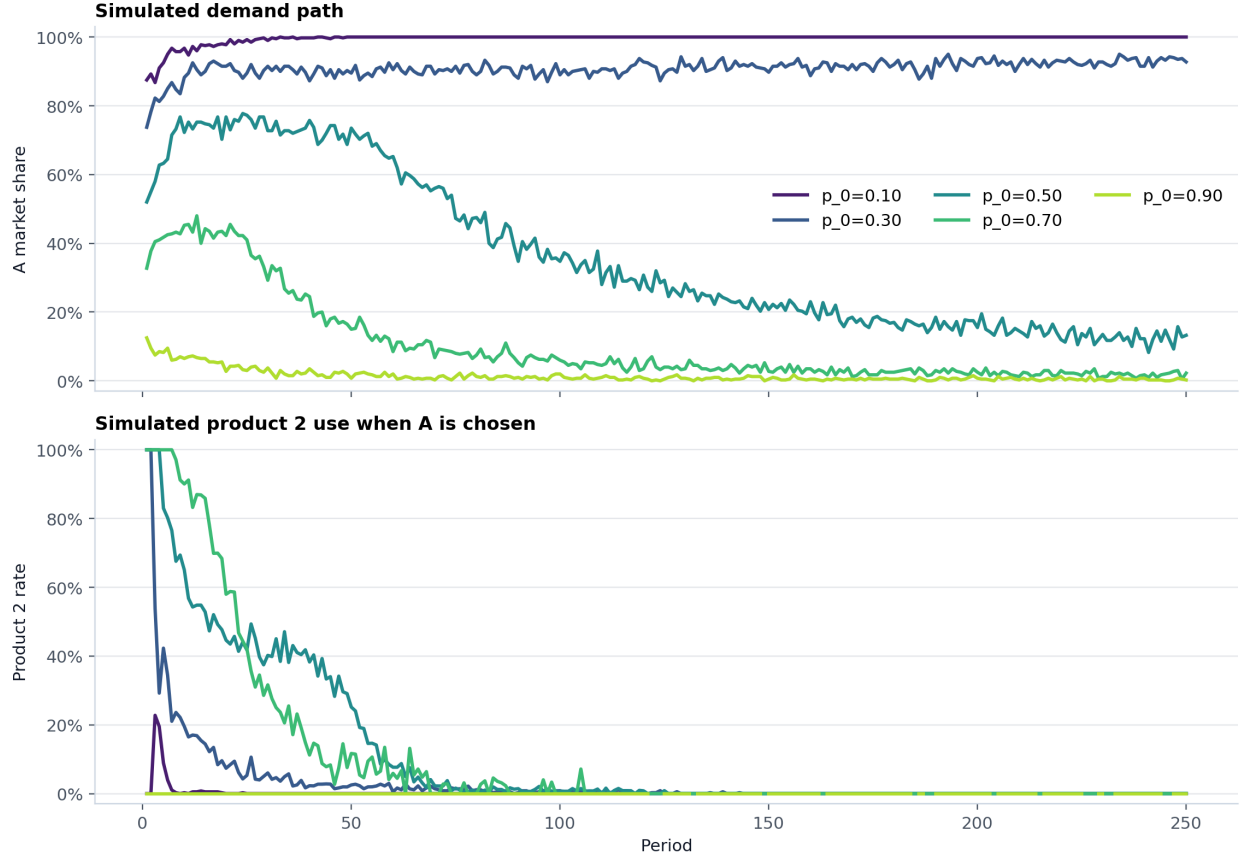


Figure 5: Simulated demand and product 2 usage paths for selected values of p_0 .

Figure 5 shows simulated histories under the solved best-response policy. The top panel tracks Seller A's market share over time. The bottom panel tracks the probability that Seller A uses product 2 conditional on being chosen.

As p_0 increases, Seller A is chosen less often. Conditional on being chosen, product 2 use tends to rise over the intermediate range because reputation investment becomes more important. At very high p_0 , demand is so low that investment is often not worthwhile.